MARKING SCHEME

Model Paper — 05

Third Term Test (Full Syllabus)

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Note (for teachers):

- Part A: 2 marks for each question (generally 1 mark for the correct method and 1 mark for the correct final answer). — Total 40 marks
- Part B: 10 marks for each question. — Total 60 marks. The marks for each step are shown in the right-hand column.
- Award full marks for correct alternative methods. Accept a reasonable range for values read from graphs.

Part A — Answers

1. $\sqrt{0.0196} = \sqrt{\frac{196}{10000}} = \frac{14}{100}$

Final answer: 0.14

(2 marks)

2. $\frac{5}{2} - \frac{4}{3} = \frac{7}{6}$; $\frac{7}{6} \div \frac{7}{2} = \frac{7}{6} \times \frac{2}{7}$

Final answer: $\frac{1}{3}$

(2 marks)

3. $(x^2 - 25) - (x^2 - 4x + 4)$

Final answer: $4x - 29$

(2 marks)

4. Product 28, sum $-11 \Rightarrow -4, -7$

Final answer: $(x - 4)(x - 7)$

(2 marks)

5. $3a(a^2 - 4)$

Final answer: $3a(a - 2)(a + 2)$

(2 marks)

6. $\frac{x(x - 4)}{(x - 4)(x + 4)}$

Final answer: $\frac{x}{x + 4}$

(2 marks)

7. Arc length = $\frac{45}{360} \times 2 \times \frac{22}{7} \times 14 = 11$ cm ; Perimeter = 11 + 14 + 14

Final answer: 39 cm

(2 marks)

8. $Q\hat{P}R = x \Rightarrow P\hat{Q}R = P\hat{R}Q = 2x$; $x + 2x + 2x = 180°$

Final answer: 36°

(2 marks)

9. $\log_2 4x = 5 \Rightarrow 4x = 2^5 = 32$

Final answer: x = 8

(2 marks)

10. $\lg 0.03 = \lg 3 - \lg 100 = 0.4771 - 2$

Final answer: $\bar{2}.4771$

(2 marks)

11. $T_n = 40 - 3(n - 1) < 0 \Rightarrow n - 1 > 13.3 \Rightarrow n = 15$ ($T_{14} = 1$, $T_{15} = -2$)

Final answer: the 15th term

(2 marks)

12. $B \subset A \Rightarrow A \cap B = B$, $A \cup B = A$

Final answer: n(A ∩ B) = 12 ; n(A ∪ B) = 20

(2 marks)

13. 20 000 × 1.05 = 21 000 ; 21 000 × 1.05 = 22 050 ; 22 050 − 20 000

Final answer: Rs. 2 050

(2 marks)

14. Man-hours = 10 × 8 × 12 = 960 ; 960 ÷ (12 × 10)

Final answer: 8 days

(2 marks)

15. 72 km/h = $72 \times \frac{1000}{3600} = 20$ m/s ; 200 ÷ 20

Final answer: 10 seconds

(2 marks)

16. $y(x - 1) = 2x + 3 \Rightarrow xy - 2x = y + 3 \Rightarrow x(y - 2) = y + 3$

Final answer: $x = \frac{y + 3}{y - 2}$

(2 marks)

17. Number of balls that are not white = 3 + 2 = 5 ; $\frac{5}{10}$

Final answer: $\frac{1}{2}$

(2 marks)

18. $3 + 7 + x + 10 + 12 = 5 \times 8 = 40 \Rightarrow 32 + x = 40$

Final answer: $x = 8$

(2 marks)

19. $B\hat{A}D = 180° - (35° + 60°) = 85°$; $B\hat{C}D = B\hat{A}D$ (opposite angles)

Final answer: 85°

(2 marks)

20. $2 \times \frac{22}{7} \times 7 \times h = 440 \Rightarrow 44h = 440$

Final answer: 10 cm

(2 marks)

Part B — Answers and Mark Allocation

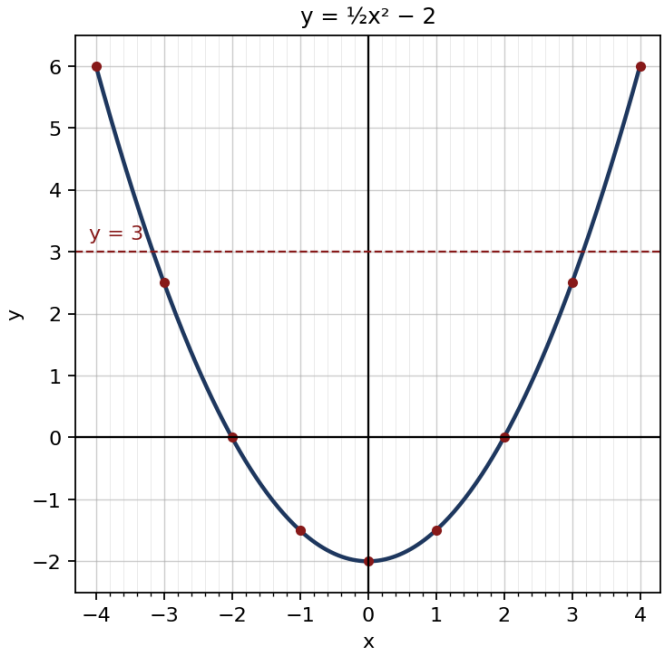
Question 1

(a)	
$x^2 - x - 6 = (x - 3)(x + 2)$; common denominator $(x - 3)(x + 2)$	**1**
$\frac{3 + (x - 3)}{(x - 3)(x + 2)}$	**1**
$= \frac{x}{(x - 3)(x + 2)}$	**1**
(b)	
$2p + 3b = 410$ .....①	**1**
$3p + 2b = 340$ .....②	**1**
① + ②: $p + b = 150$; ① − ②: $b - p = 70$	**1**
$b = 110$, $p = 40 \Rightarrow$ pen Rs. 40, book Rs. 110	**1**
(c)	
$x^2 + x - 6 = 14 \Rightarrow x^2 + x - 20 = 0$	**1**
$(x + 5)(x - 4) = 0$	**1**
$x = 4$ or $x = -5$	**1**

Total marks: 10

Question 2

(a)	
$x = -3 \Rightarrow y = 2.5$; $x = 1 \Rightarrow y = -1.5$	**1**
(b)	
Correct scale and axes	**1**
Plotting the 9 points correctly	**1**
Smooth curve	**1**



(c)	
(i) Minimum value −2 ; turning point (0, −2)	**1**
(ii) $x = -2$ or $x = 2$	**1**
(iii) Drawing the line $y = 3$	**1**
$x \approx -3.2$ or $x \approx 3.2$ (accept ±3.1 – ±3.2)	**1**
(iv) $-3 \leq x \leq 3$	**1**
(d)	
$y = \frac{1}{2}x^2$	**1**

Total marks: 10

Question 3

(a)	
$\frac{90}{360} \times \frac{22}{7} \times 14 \times 14$	**1**
$= 154\ m^2$	**1**
(b)	
Rectangle $= 28 \times 14 = 392\ m^2$	**1**
Shaded region $= 392 - 2 \times 154 = 84\ m^2$	**1**
(c)	
Length of one arc $= \frac{90}{360} \times 2 \times \frac{22}{7} \times 14$	**1**
$= 22$ m	**1**
Perimeter = DC + the two arcs = 28 + 22 + 22 = 72 m	**1**
(d)	
Grass: 308 × 200 = Rs. 61 600	**1**
Paving: 84 × 1 000 = Rs. 84 000	**1**
Total cost = Rs. 145 600	**1**

Total marks: 10

Question 4

(a)	
DÂE = EÂB (AE is the angle bisector)	**1**
EÂB = AÊD (AB ∥ DC, alternate angles)	**1**
∴ DÂE = AÊD ⇒ AD = DE ; ADE is an isosceles triangle	**1**
(b)	
DE = AD = 7 cm ; DC = AB = 12 cm	**1**
EC = 12 − 7 = 5 cm	**1**
(c)	
DÂB + AB̂C = 180° (co-interior angles, AD ∥ BC)	**1**
½ DÂB + ½ AB̂C = 90° ⇒ the sum of two angles of the triangle formed by AE and BF is 90°	**1**
Remaining angle = 180° − 90° = 90° ⇒ AE ⊥ BF	**1**
(d)	
Similarly, BCF is isosceles ⇒ CF = BC = 7 cm ⇒ DF = 12 − 7 = 5 cm	**1**
EF = DE − DF = 7 − 5 = 2 cm	**1**

Total marks: 10

Question 5

(a)	
Discount = 120 000 × $\frac{12}{100}$ = Rs. 14 400	**1**
Amount to be paid = Rs. 105 600	**1**
(b)	
Initial payment = Rs. 30 000 ; instalments = 12 × 8 500 = Rs. 102 000	**1**
Total payment = Rs. 132 000 ⇒ extra = Rs. 12 000	**1**
$\frac{12\,000}{120\,000}$ × 100 = 10%	**1**
(c)	
Profit = 105 600 − 88 000 = Rs. 17 600	**1**
$\frac{17\,600}{88\,000}$ × 100 = 20%	**1**
(d)	
Required selling price = 88 000 × $\frac{125}{100}$ = Rs. 110 000	**1**
Marked price M ⇒ 0.88M = 110 000	**1**
M = Rs. 125 000	**1**

Total marks: 10

Question 6

(a)	
$\frac{1}{6} + \frac{1}{4}$	**1**
$= \frac{5}{12}$	**1**
(b)	
Time = $\frac{12}{5}$ hours	**1**
= 2 hours 24 minutes	**1**
(c)	
$\frac{1}{6} + \frac{1}{4} - \frac{1}{12}$	**1**
$= \frac{4}{12} = \frac{1}{3}$	**1**
Time = 3 hours	**1**
(d)	
Part filled in 1 hour = $\frac{5}{12} \Rightarrow$ remainder = $\frac{7}{12}$	**1**
A and C together (per hour) = $\frac{1}{6} - \frac{1}{12} = \frac{1}{12}$	**1**
Time = $\frac{7}{12} \div \frac{1}{12}$ = 7 hours	**1**

Total marks: 10